

# Grounded Local RAG

One-page engineering explanation | FastAPI + local vector store | 18 September 2026

## 1. Design parameter: chunk size

I chose a **180-word target with a 40-word overlap**, but changed fixed word windows to sentence-aware windows. The target keeps a short factual procedure together; overlap protects a boundary fact. Preserving sentence boundaries matters because support is now verified at sentence level. The strategy and parameters are persisted with every document, and chunks remain inspectable through the API.

## 2. Failure observed while building

An early gate unioned terms across a retrieved chunk. "Does the safety officer inspect battery cabinets?" incorrectly returned grounded: `true`: one sentence mentioned the officer and another cabinet inspection, but no sentence established that relationship. Aggregate term coverage masqueraded as support. I fixed it by requiring one sentence to cover at least **60%** of a one-part question; the trace now exposes the failed span check rather than inventing a connection.

## 3. Metric tracked: local query latency

I added a fixed, manually-authored **15-case** regression set across TXT, a two-page PDF, direct and multi-fact questions, an explicit negative fact, and three unsupported/adversarial questions. Final local-hash + hybrid dense/BM25 run: **Recall@1=1.00**, Recall@3=1.00, fact/citation pass=1.00, unsupported abstention=1.00, median latency=**7.60 ms**, p95=11.09 ms. This is a small regression result, **not** a production accuracy claim.

## 4. Not finished and next step

The offline fallback is still hashed lexical embeddings, not a semantic local model; the compact suite is for regression rather than independently collected at scale. I did not add OCR, tables/layout-aware parsing, metadata filtering, authentication, or multi-user writes. Next I would collect a held-out human-authored corpus, compare OpenAI embeddings with a local sentence-transformer using Recall@k and citation faithfulness, then tune the 60% evidence threshold only on a separate development split.

Honest scope: runnable offline by default; hybrid ranking and sentence-level evidence are inspectable; OpenAI calls remain optional and guarded.